SonicAtlas - 20 likely viva questions (Team 05)

1. What is the project in one line? SonicAtlas explores how Spotify audio features and lyrics differ across genres and eras, and gives a website to query them in the browser.

2. Where does the data come from? Three public Kaggle datasets: Spotify 1.2M+ songs, Spotify tracks with 114 genres, and Genius lyrics. The live Spotify API is closed for new apps.

3. What is MICE and why did you use it? Multiple Imputation by Chained Equations: each column with gaps is predicted from the others, in turns. We used it for the impossible 0 BPM tempos.

4. How did you know MICE worked? We hid 5% of known values, imputed them, and compared the error with mean filling. MICE was better for energy, valence and tempo.

5. Why log transform duration? Duration is strongly right-skewed (G1 about 8). log1p brings G1 to about 0.2, so distance methods are not dominated by very long tracks.

6. What are G1 and G2? G1 is the adjusted Fisher-Pearson skewness; G2 is the bias-corrected excess kurtosis. Pearson kurtosis is G2 plus 3.

7. Why Tukey quartiles? It is the method used in our tutorials: split the data at the median and take the median of each half. We also show the pandas value; on a million rows they agree.

8. Why not delete outliers? Many are real: spoken word, ambient, very fast songs. We flag them (Tukey fences and Isolation Forest) so users can include or exclude them.

9. Pearson vs Spearman? Pearson measures linear relation, Spearman monotonic relation using ranks. Both agree here, so outliers are not driving the results.

10. PCA vs t-SNE vs LLE? PCA is linear, fast and gives loadings; t-SNE keeps local neighbours best (trustworthiness 0.98); LLE unfolds smooth manifolds and was weakest (0.67).

11. How are PCA and SVD related? PCA is the SVD of the centred data matrix: the squared singular values give the variances. We checked this numerically.

12. What is trustworthiness? It measures whether points that are neighbours in the map were also neighbours in the original space (1 is perfect).

13. Why is the genre silhouette negative? Genres overlap strongly in audio space; genre is partly a cultural label, so no projection separates them cleanly.

14. What is STL? Seasonal-Trend decomposition using Loess: it splits a time series into trend, seasonal and remainder. We used it on monthly release means.

15. What did you find about loudness? Mean loudness rose from -17.4 to -9.7 dB between 1959 and 2020, with change points in 1989 and 1999.

16. What does the ANOVA say? $F = 512.1$, $p < 0.001$: danceability differs across genres. Eta squared = 0.425, a large effect. Tukey HSD says which pairs differ.

17. Why also Kruskal-Wallis? Danceability is bounded and variances are unequal (Levene), so we confirmed with a rank-based test. It agrees.

18. What does the chi-square test show? Mode (major/minor) depends on genre ($V = 0.33$, moderate) but only negligibly on decade ($V = 0.05$).

19. Why DuckDB-WASM? It runs SQL on Parquet inside the browser: fast, private, no server and free hosting on GitHub Pages.

20. What are the limitations? Spotify features are model-made; recent years dominate; only 10.7% of genre tracks join to the audio table; lyrics are English-only; the 2020s have one year; the paper is not yet submitted.